

Lesson 2 — Build the Model & Sequencing

Matches LEGO Teacher Portal lesson: "Robot Food Service".

Objective (I can...): I can build the unit model and order instructions so the machine performs a multi-step sequence reliably. **CS concept:** Sequencing and execution order (statements run top to bottom; order changes outcome) · **Builds on:** Lesson 1 (running a program).

AI Connection (2 min): A voice assistant runs an ordered pipeline: capture audio → transcribe → interpret → respond. *First-principles question:* "Which two of those steps absolutely cannot be swapped, and why?"

Materials: LEGO CS & AI kit + device per pair, notebooks. **This is the unit's build lesson** — build the Basics model now and keep it assembled through Lesson 5.

60-minute flow - Warm-up (8): Sequence sabotage — pairs get a 6-step process written out of order and reorder it, then identify which single swap would be most destructive. Connect to execution order. -

Concept + demo (10): Run the same block set in two different orders; show the different outcomes. Introduce reading a program top-to-bottom as "tracing" and model tracing one line at a time out loud. -

Build/Code (30): Pairs **build the Basics model** per the Canvas instructions (the one build for the unit), then write and run a 5-step sequence (e.g., light on → drive forward → turn → play tone → stop), tracing each step before running. *Swap at 15 min.* - **Challenge (7):** Reorder the same blocks into a deliberately different, sensible behavior; record both the old and new trace in the notebook. - **Share + pack-up (5):** Each pair shows original vs. reordered program and names what the order changed. **Leave the model built** on its tray.

Pair roles: During the build, Driver connects bricks while Navigator reads and checks steps; they swap for coding. *Swap at 15 min. (Trio rotates build/read/test.)*

Differentiation: *Support* — a worked 5-step example to adapt, not copy. *Extension / fast-finisher* — write an 8-step sequence with two outputs running close together. *Reteach* — trace the program aloud, one line at a time, running after each line.

Exit check: Pair produces a working 5-step sequence and can trace it, explaining why order matters. **At-home (optional):** Write the exact start-up sequence for a device you use (game console, microwave); note what breaks if two steps swap. ~10 min.
